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EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Apr. 2019 – Jul. 2024

BSc in Computer Science and Engineering

Dhaka, Bangladesh

- **CGPA: 3.91** ; Dean's List Award
- Notable Courses: Machine Learning, Compiler, Computer Networking, Security, AI, Architecture, Operating Systems, Data Structure & Algorithm

RESEARCH

CompassLLM: A Multi-Agent Approach toward Geo-Spatial Reasoning for Popular Path Query

[arxiv Preprint] — Undergraduate Thesis — supervised by Mohammed Eunus Ali & Md Rizwan Parvez

- Proposed a novel multi-agent framework to tackle geo-spatial reasoning problems, particularly identifying and generating popular paths from historical trajectory data
- Orchestrated the agents in two stages – a *Search* stage for extracting popular paths and a *Generate* stage for synthesizing novel routes
- Evaluated using GPT-4o, GPT-o3-mini and Llama 3.1 8b and demonstrated superior performance on real and synthetic datasets, especially in sparse data scenarios where other methods underperform
- Submitted to EACL 2026. Implementation to be open-sourced

EXPERIENCE

Software Engineer

Mar. 2024 – Present

Zelf AI

Brooklyn, NY, USA

- Building data pipelines collecting and processing 80M+ daily video analytics from social media public APIs
- Developing automation tools serving 100+ enterprise clients
- Collaborating with retail partners and cross-functional teams for critical features
- **Tech Stack:** Python, Selenium, Playwright, Appium, Burp Suite, Mitmproxy, ADB, GCP, Docker, GitHub Actions, Terraform

Software Consultant

Jun. 2025 – Present

Shottify

Dhaka, Bangladesh

- Conducted sessions on Information Retrieval and Social Media Analysis
- Developed News Aggregator for renowned international news portals including NYT, Reuters - [See More]

Research Assistant

Jul. 2022 – Jun. 2024

Institute of Water and Flood Management, BUET

Dhaka, Bangladesh

- Practical implementation of an Early Warning System for River Erosion prone areas in Bangladesh
- GIS Data Visualization using Google Maps, React & Express - [Live Here]
- Published in *Frontiers: A Riverbank Erosion Early Warning System*

Machine Learning Intern

May – Jun. 2023

Era-InfoTech Ltd

Dhaka, Bangladesh

- Data analytics using Unsupervised Learning (KNN) and NLP on real business data

PROJECTS

Online News Scraper | *Python, FastAPI*

- News Aggregator for renowned international news portals including NYT, Reuters

Stock Trader | *ML, Python*

- Trained a rule-based stock trading agent on DSE time-series data; yielded 18% ROI in simulated backtests

Maps 'n Bags | *ExpressJS, PostgreSQL, Firebase, Selenium*

- Developed an itinerary planner with custom trip builders, suggestions, travel tracking & budgeting with synced data & distances via Google Maps API

Creative Production Management | *System Design, BPMN, UI Mockups*

- Designed an end-to-end production workflow for a creative agency — covering quotation, delegation, man-hour tracking, and final delivery

Withered Away | *ExpressJS, OracleDB*

- Built dashboard to manage residents, amenities, and room allocations for retirement home officials, doctors, and members

SCORES & ACHIEVEMENTS

1st / 400 – Zelf Hackathon 1.0 Feb. 2024
Built an automation platform that solved long-standing engineering backlogs; led to recruitment interview
5th / 68 – IEEE GameJam Jan. 2023
Developed an Asteroids game using p5.js [Play Now] - don't forget to break the fourth wall

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, Bash, x86 Assembly
Full-stack: Flask, FastAPI, React, Express, NestJS, Apollo, Prisma
DevOps: GCP, Git, Docker, Kubernetes, Terraform, Github Actions, Firebase, Linux Shell
Testing & Automation: Selenium, Playwright, Appium, Burp, ADB
Libraries & ML: NumPy, Pandas, LangChain, OpenCV, PyTorch, scikit-learn, NLTK, Matplotlib, Seaborn
Databases: PostgreSQL, MongoDB, MySQL, Oracle

EXTRACURRICULAR

Director of Marketing & Communication Jun. 2023 – Apr. 2024
BUET Cyber Security Club Dhaka, Bangladesh

- Promoted cybersecurity events & hosted inter-university CTF (Catch The Flag) contests

Organizer Jun. – Jul. 2023
BUET CSE Fest 2023 Dhaka, Bangladesh

- A proud part of the organizing team of the 2023 edition of the annual departmental festival consisting inter-university competitions such as hackathon, programming contest, deep learning competition, CTF competition and extra-curricular events such as cultural program, dance mob, treasure hunt, indoor and outdoor games

President of Writing & Publication Feb. 2018 – Feb. 2019
Notre Dame English Club Dhaka, Bangladesh